

Writing Equations of Lines

Directions: Show all work. No calculators.

1. Write the equation of the line through $(0, -4)$ and $(6, 5)$ in slope-intercept form.
2. Write the equation of the line through $(0, 7)$ and $(-4, -1)$ in slope-intercept form.
3. Write the equation of the line through $(0, -3)$ and $(5, -8)$ in slope-intercept form.

4. Write the equation of the line through $(0,2)$ and $(-6,5)$ in slope-intercept form.

5. Write a point-slope equation for the line with slope $-\frac{2}{3}$ through $(6, -1)$.

6. Write a point-slope equation for the line with slope $\frac{5}{2}$ through $(-4,7)$.

7. Write a point-slope equation for the line with slope -1 through $(3, -5)$.

8. Write a point-slope equation for the line with slope 4 through $(-3, 2)$.
9. A line goes through the points $(-5, -7)$, $(-2, -1)$, $(1, 5)$, $(4, 11)$. Write four different point-slope equations for the line.
10. A line goes through the points $(-5, 8)$, $(-3, 5)$, $(-1, 2)$, $(1, -1)$. Write four different point-slope equations for the line.
11. A line goes through the points $(-8, -4)$, $(-5, -3)$, $(-2, -2)$, $(1, -1)$. Write four different point-slope equations for the line.

12. A line goes through the points $(-6,9)$, $(-3,3)$, $(2,-7)$, $(5,-13)$. Write four different point-slope equations for the line.
13. Write an equation of the line through $(-3,4)$ and $(5,8)$ in point-slope form.
14. Write an equation of the line through $(2,-5)$ and $(6,7)$ in point-slope form.
15. Write an equation of the line through $(-4,-2)$ and $(2,-5)$ in point-slope form.
16. Write an equation of the line through $(1,6)$ and $(7,-3)$ in point-slope form.